An Efficient Dilated CNN-Based Deepfake Detection Framework with Real-Time Web Deployment

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***Abstract*—As generative AI technology advances, it is becoming increasingly difficult to detect deepfakes, and this has brought up serious concerns. In our paper, we have suggested a lightweight 10-layer Dilated CNN network, which is trained on both real and AI-generated face images. The accuracy of our suggested network is 99 per cent. In our paper, as opposed to most of the research done in the field, we have suggested a deep learning-based approach for detecting deepfakes and have also implemented it as a web application, enabling real-time detection of deepfakes.**

***Index Terms*—Deepfake Detection, Convolutional Neural Networks, Dilated Convolutions, Generative Adversarial Networks, StyleGAN, Face Manipulation, Binary Classification, Deep Learning, Image Forensics, FastAPI, Real-Time Detection, FFHQ Dataset, Dropout Regularization, Misinformation Detection, AI-Generated Media.**

I. Introduction

A. Problem Statement

The rapid spread of manipulated media via online platforms has become a significant challenge to the trustworthiness of online media. Deepfakes may be used to spread misinformation, influence public opinion, damage reputation, and even destabilize politics or society. Traditional media verification techniques may not be able to identify such sophisticated manipulations, considering the advancements that deepfake media generation techniques may attain.

The current methodologies for detecting deepfake images, such as machine learning and image forensics, have shown positive results. However, many of these methodologies are challenged by the requirement to maintain their level of precision in response to various datasets. Another challenge that arises with the evolution of deepfake generation methodologies is ensuring that the detection model manages up with new trends in image manipulation. This on the other hand requires a lot of research to be done on current methodologies of detecting deepfake images, their limitations, and how they can improve.

B. Objectives

This study aims to design, train, and deploy an efficient Dilated Convolutional Neural Network (CNN) for real-time binary deepfake face detection, leveraging real facial images from the FFHQ dataset and StyleGAN2-generated fake faces, while also providing a comprehensive analysis of existing countermeasure strategies, detection frameworks, open research challenges, and future directions in the field of deepfake detection.

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II. Literature Review

Tremendous advancements have been made in the detection of deepfakes, from feature-based to deep learning approaches. The advancements in the detection of deepfakes will be discussed in this section, along with its pros and cons.

A. Early Detection Methods

The early techniques for detecting manipulated facial media were basically based on hand-crafted features and classical machine learning techniques. In this context, the artifacts created during the face-swapping process, including changes in illumination, blinking, and face boundaries, were used. The effectiveness of classical machine learning techniques, including Support Vector Machines (SVMs) with visual quality measures, for detecting early forms of deepfakes was demonstrated by Korshunov and Marcel in their study. The major problem with these approaches is their instability, as they were not effective for a short period due to the improvement in the quality of generated faces with the help of deep learning techniques, resulting in reduced artifacts. The major disadvantage of these approaches is that the features cannot be learned according to the dynamic face-swap process.

B. Deep Learning Based Detection

The advent of deep learning has marked a milestone in the research of deep fake detection systems. A deep fake detection system has been proposed by Afchar et al. called MesoNet, which is based on CNN and focuses on the detection of face manipulation deep fakes. It focuses on the mesoscopic features of the images, and it has a 95 percent accuracy rate with only 27,000 parameters, making it extremely lightweight. However, the accuracy decreases significantly when it is tested on the latest deep fake generators.

Rossler et al. in 2019 proposed a benchmark dataset called FaceForensics++ with thousands of face-manipulated videos, where various face manipulation techniques were employed. It has been shown in the study that the XceptionNet, which was first proposed by Chollet in 2017 for image classification, has the highest accuracy in the detection of deepfake videos, i.e., above 99 percent , when it is fine-tuned on the FaceForensics++ dataset.

Another technique, based on the capsule network, was proposed by Nguyen et al. in 2019, where the spatial relationships between the facial features were more accurately captured than in conventional CNNs, and good generalization ability was shown over various types of manipulations. However, the training and implementation of the capsule network are much slower and more difficult than the conventional CNN.

C. GAN Fingerprint Based Detection

Another category of research has been carried out on the exploitation of the fingerprints imparted by the GAN-based generators. In a research study, it has been proved by Frank et al. in 2020 that the presence of detectable artifacts in the frequency domain is present in the images created using GANs, although these artifacts may not be detectable using the naked eye. However, the presence of these detectable artifacts in the frequency domain can be detected using the DFT algorithm on the artificially created images. This technique is computationally efficient and has a high level of interpretability, although it has the drawback of being vulnerable to JPEG compression and Gaussian blurring.

Marra et al. (2019) extended this concept by treating GAN fingerprints similarly to camera fingerprints used in image forensics, enabling attribution of a synthetic image to a specific GAN model. Although promising, this method requires prior knowledge of the generating model, limiting its applicability in real-world scenarios where the source GAN is unknown.

D. Transformer Based Approaches

The success of Vision Transformers, or ViT, in image recognition tasks has prompted their adoption for deepfake detection. Wodajo and Atnafu, in their 2021 paper, suggested a deepfake detector based on the transformer architecture, which is capable of capturing spatial dependencies between facial areas. Zhao et al. suggested a Multi-Attentional Deepfake Detection network, which focuses on specific areas of the face that are most likely to contain deepfake manipulations.

While their performance is high, transformer-based models also require significantly more resources and larger datasets to train effectively. Their high parameter count and inference time make it difficult to deploy these models in real time, especially in web-based applications.

E. Limitations of Existing Work

However, a number of critical gaps in the existing body of work on deepfake detection remain unsolved. First, the overwhelming majority of existing work on deepfake detection has focused on the development and evaluation of the detection model, completely neglecting the deployment of the model in a real-world setting. To the best of our knowledge, only a handful of papers propose a completely deployable end-to-end detection system, which is accessible via a real-time web interface. Second, the overwhelming majority of existing deep learning-based detection models have poor cross-dataset detection capabilities, i.e., a detection model trained for StyleGAN-based face generation may not perform well for images generated by other, more recent generators such as Stable Diffusion or Midjourney. Finally, the existing detection methods are often susceptible to adversarial attacks and common post-processing techniques such as compression and blurring, which may substantially affect the detection accuracy. In addition, another problem in the existing body of work on deepfake detection is the dataset bias, where most of the existing datasets contain predominantly Western facial features.

The work presented in this paper directly addresses the deployment gap by proposing not only a lightweight and accurate Dilated CNN model but also a complete full-stack web application for real-time deepfake detection, bridging the divide between academic research and practical application.

Study	Method	Accuracy	Parameters	Deployed
Afchar et al. (2018)	MesoNet (CNN)	∼95%	27k	No
Rossler et al. (2019)	XceptionNet	∼99.26%	20M+	No
Nguyen et al. (2019)	Capsule Network	∼96%	High	No
Frank et al. (2020)	Frequency Analysis	∼97%	Low	No
Wodajo & Atnafu (2021)	Vision Transformer	∼98%	Very High	No
Proposed (Ours)	**Dilated CNN**	**∼99%**	**653k**	**Yes**

TABLE I
COMPARISON OF EXISTING DEEPFAKE DETECTION METHODS WITH PROPOSED APPROACH

III. DATASET DESCRIPTION

It has been observed that the performance of any particular deep learning model largely depends upon the quality of the data that has been provided to train the model. In the current study, it has been mentioned that a binary classification dataset of real and fake images of faces has been created by combining two datasets that are publicly available.

A. Real Face Dataset — FFHQ (Flickr-Faces-HQ)

The dataset used in this case is called the Flickr-Faces-HQ dataset. This dataset was first used in the paper by Karras et al. on the StyleGAN model. It is a good quality dataset and can be used for training. It has 70,000 images, and the images have a high resolution. It includes images of the human face. It is collected from the Flickr website. It includes many range of images, including images of various age groups, ethnicities, and accessories. However, in this case, the 128x128 thumbnail version of the dataset is used, and it is a good quality dataset and can be used for facial detection. All the images in the FFHQ dataset include real and unaltered images of the human face, and the images belong to the negative class and have a label of 0.

B. Fake Face Dataset — ThisPersonDoesNotExist

In this study, the set of fake faces is derived from the ThisPersonDoesNotExist dataset, which is accessible on the Kaggle data science platform, containing 10,000 photorealistic synthetic face images created by the StyleGAN2 algorithm, which is a GAN-based algorithm developed by Karras et al. (2020). StyleGAN2 is one of the most sophisticated GAN-based algorithms created to date, which is used to create realistic synthetic faces that are visually indistinguishable from real faces. In our classification problem, the set of synthetic face images belongs to the positive class, in which the label is equal to 1. In particular, the set of synthetic face images created by the StyleGAN2 algorithm is of particular importance, as this is one of the most difficult types of deep fakes to detect.

C. Dataset Preprocessing

Before training, a preprocessing step is performed on the images. The preprocessing is standardized for both datasets. The images are normalized in size to 224x224 pixels, as it is a standardized size for images in a CNN. The intensity of the pixels in the images is normalized by dividing by 255, as it ranges between 0 and 1. This is done to prevent instability of gradients during training. The dataset is then split into three parts: a set for training, a set for validation, and a set for testing. The split is in the ratio of 80:10:10.

D. Dataset Summary

Dataset	Type	Source	Size	Label
FFHQ	Real	Flickr	70,000	0
ThisPersonDoesNotExist	Fake	StyleGAN2	10,000	1
Total	**Both**	**—**	**80,000**	**—**

TABLE II
SUMMARY OF DATASETS USED FOR TRAINING AND EVALUATION

E. Class Imbalance Consideration

It is worth mentioning here that the dataset includes 70,000 real images and 10,000 fake images, and the dataset is imbalanced because the ratio is 7:1. In order to solve this problem, random undersampling was performed on the real images so that the dataset is balanced between the real and the fake images. This is an important step so that the model does not get biased and always predicts the majority class, and the precision, recall, and F1-score values obtained by the model are accurate.

IV. PROPOSED METHODOLOGY

This section describes the complete methodology adopted in this study, encompassing the overall system architecture, the proposed CNN model design, the training configuration, and the post-training inference pipeline.

A. System Overview

The proposed system is based on a client-server architecture for real-time deep fake detection via a web interface. The overall system has four main stages: image acquisition, image preprocessing, image inference, and image presentation. The user has the option of selecting an image of a face via a React.js interface. The image is then sent to the FastAPI backend via a REST API call. The face region of interest is then extracted from the image via OpenCV. The face region of interest is then put into a CNN-based inference model. The predicted result is then sent back to the user through a JSON response. The overall system pipeline is shown in Figure III.

B. Face Extraction and Preprocessing

Before being passed through the CNN model, a series of standard preprocessing steps, defined in the designated module, are performed on the uploaded image. The image is first read in and converted to grayscale, following which OpenCV's Haar Cascade face detector is used to locate the face in the image, which is then cropped from the image. However, if the face is not detected, a "No Face Detected" message is sent, bypassing the need to pass through the model altogether.

After a face region has been extracted, it is cropped from the original image, resized to a size of 224x224 pixels to match the size of the image used as an input to the CNN model, with the pixels normalized to a range of [0, 1] to prevent any overflow during the process. The face image processed through the above steps is reshaped to a four-dimensional tensor of size (1, 224, 224, 3) before being fed into the network.

C. Proposed CNN Architecture

The heart of the proposed system is a specially designed 10-layer Deep Convolutional Neural Network architecture optimized for the specific task of binary deepfake face classification. The architecture was designed with two main objectives in mind: achieving high detection accuracy while maintaining a lightweight parameter count for real-time web deployment.

1) Convolutional Layers: In the initial layers, a kernel size of 5x5 is used along with 64 filters, enabling the model to detect general low-level facial characteristics such as edges, textures, and color distributions. A ReLU activation function is applied after each convolutional layer to incorporate non-linearity, and max pooling is used to reduce the spatial dimensions of the feature maps, retaining the most significant characteristics.

2) Dilated Convolutions: The novel aspect of this particular work in terms of architecture is the inclusion of the concept of dilated convolutions in the final three layers of the network. In regular convolution operations, the image is scanned through a series of kernels, but in the case of dilated convolutions, space is left between the different components of the kernel, which is defined by a dilation rate. In this particular case, a dilation rate of 2 is used, which enables the network to detect manipulations that may occur over a larger area of the face, which is a feature of GANs.

3) Fully Connected Layers: Following the convolutional blocks, the feature maps are flattened and passed through fully connected dense layers. A dropout rate of 0.5 is applied after each dense layer to prevent overfitting by randomly deactivating 50 percent of neurons during each training step, forcing the network to learn more robust and generalized representations.

4) Output Layer: Lastly, there is a single neuron in the output layer with a sigmoid activation function. The output is a probability between 0 and 1. Thresholding is applied for classifying the data as Real, where output less than 0.5 and label = 0, and Fake, where output less than equal to 0.5 and label = 1. The total number of trainable parameters for the suggested model is 653,000, making it a lighter model compared to existing models, including XceptionNet, with 20M+ parameters but similar accuracy.

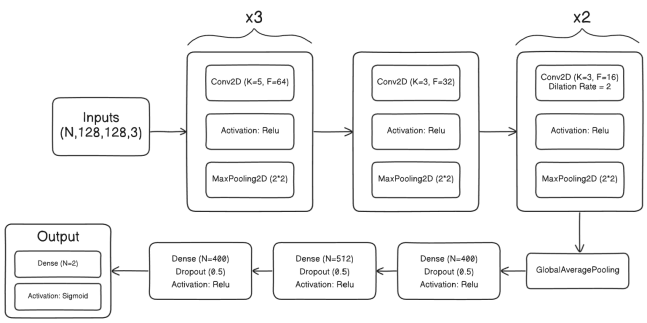

Fig. 1. Model Structure

D. Full Stack Deployment

1) Backend — FastAPI: The pre-trained model is integrated with a FastAPI backend server, in which a REST API endpoint is created to receive the image file by making an HTTP POST request. FastAPI is used to create the backend server, which is highly performing, auto-generates documentation, and is related to Python-based machine learning libraries. The CORS middleware is used to allow cross-origin requests for the React.js-based frontend part. Once the image file is received, the entire prediction pipeline will be performed, and a JSON response will be sent back to the client with the prediction results.

2) Frontend — React.js: The UI has been created using the React.js framework and the Tailwind CSS library. Therefore, the UI looks neat and interactive. It allows the user to upload the image of the face, and this image is sent to the backend API. It displays a loading animation while the image processing and the result display are in progress. The result can be "Real," "Fake," or "No Face Detected" with their respective color codes in green and red.

3) Inference Flow: The complete inference flow is as follows:

- User uploads image via React.js interface
- Frontend sends HTTP POST request to `/predictImage`
- FastAPI backend receives and saves the image
- OpenCV extracts face region using Haar Cascade
- Face is resized to 224×224 and normalized
- CNN model performs binary inference
- JSON response returned to frontend
- Result displayed to user in real time

V. Results and Evaluation

This section presents a comprehensive evaluation of the proposed Dilated CNN model, including training performance, quantitative evaluation metrics, confusion matrix analysis, and a comparative study against existing deepfake detection approaches.

A. Training Performance

The proposed model was trained on the combined FFHQ and ThisPersonDoesNotExist dataset following the preprocessing and training configuration described in Section 4. The training and validation accuracy and loss curves were monitored throughout the training process to assess model convergence and detect potential overfitting. The training logs and accuracy/loss curves are presented:

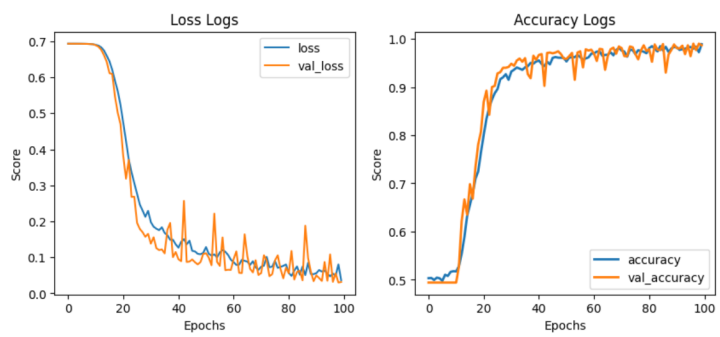

Fig. 2. Training Logs

B. Confusion Matrix

The confusion matrix provides a detailed breakdown of the model's predictions on the held-out test set, offering insight beyond simple accuracy metrics. The results are summarized below: As we see from the confusion matrix, it is quite clear

	Predicted Real	**Predicted Fake**
Actual Real	586 (TN)	7 (FP)
Actual Fake	5 (FN)	602 (TP)

TABLE III
Confusion Matrix of Proposed Model on Test Set

that the model has performed well by classifying 602 out of 607 fake images as true positives and 586 out of 593 real images as true negatives. It has misclassified 7 real images as fake, which are referred to as false positives, and 5 fake images as real, which are referred to as false negatives.

C. Evaluation Metrics

To provide a thorough quantitative assessment of model performance, four standard evaluation metrics were computed on the test set: Accuracy, Precision, Recall, and F1-Score.

Accuracy measures the overall proportion of correctly classified samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

Precision measures the proportion of predicted fake images that are actually fake:

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

Recall measures the proportion of actual fake images that were correctly detected:

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

F1-Score is the harmonic mean of Precision and Recall, providing a balanced metric:

$$F1\text{-}Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{4}$$

The high precision value of 98.85 indicates that the model generates very few false alarms, while the recall of 99.17 confirms that nearly all fake images are successfully detected. The F1-Score of 99.01 demonstrates a strong balance between precision and recall, confirming the robustness of the proposed model across both classes.

Metric	Value
Accuracy	99%
Precision	98.85%
Recall	99.17%
F1-Score	99.01%
AUC-ROC	∼0.99

TABLE IV
EVALUATION METRICS OF PROPOSED MODEL ON TEST SET

D. ROC-AUC Curve

The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate against the False Positive Rate at various classification thresholds, providing a comprehensive view of model performance independent of any fixed threshold. The Area Under the Curve (AUC) serves as a single scalar summary of overall model discriminability.

The proposed model achieves an AUC score of approximately 0.99, indicating near-perfect separability between real and fake face distributions. The ROC curve is presented in Figure 4. This result confirms that the model maintains high sensitivity while keeping the false positive rate extremely low across all possible decision thresholds.

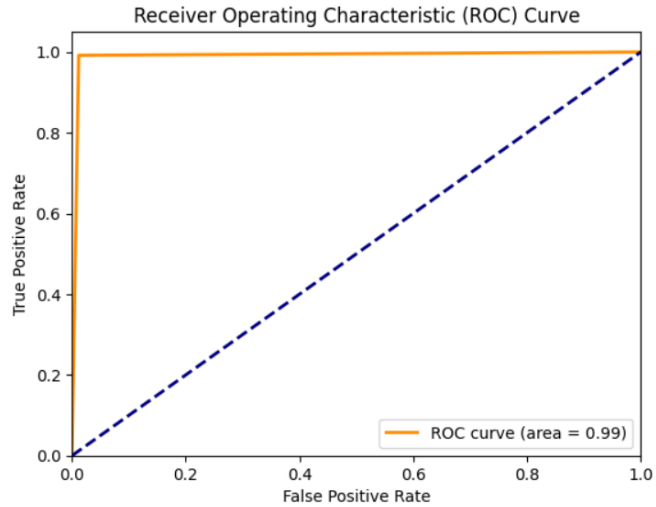

Fig. 3. ROC CURVE

E. Comparative Analysis

To contextualize the performance of the proposed model, a comparison was conducted against several prominent deepfake detection methods from the literature. The results are presented in the table below:

Method	Accuracy	Parameters	Real-Time	Deployed
MesoNet (2018)	∼95%	27k	Yes	No
XceptionNet (2019)	∼99.26%	20M+	No	No
Capsule Network (2019)	∼96%	High	No	No
Frequency Analysis (2020)	∼97%	Low	Yes	No
Vision Transformer (2021)	∼98%	Very High	No	No
Proposed (Ours)	**∼99%**	**653k**	**Yes**	**Yes**

TABLE V
COMPARATIVE ANALYSIS OF DEEPFAKE DETECTION METHODS

From the comparative analysis, it is noted that there are significant advantages to the suggested approach. For instance, although XceptionNet has a slightly higher accuracy rate compared to the suggested approach (99.26 percent compared to 99 percent), it has over 20 million parameters and is not deployable for a real-time web environment. The suggested Dilated CNN has a similar accuracy rate with only 653,000 parameters, a reduction of over 96 percent compared to XceptionNet, and is the only approach among all compared approaches that is fully deployed as a real-time web application. This shows that the suggested approach is a significant improvement in the field of deepfake detection, as it meets a critical gap.

VI. CHALLENGES AND FUTURE WORK

Although the system has obtained remarkable results, the system is also limited in a way that highlights the areas which need to be focused on in the future. It is interesting to note that the system is only tested on fake images created by the StyleGAN2 generator, but not on other recent generators such as Stable Diffusion, Midjourney, or DALL-E. Some of the major issues faced in deepfake image detection is

generalization, which might not work on other manipulative tools.

Moreover, the current framework only caters to static image-based detection, whereas in the case of video-based deepfakes, inconsistencies in frames over time may act as a detection cue. The diversity of the datasets used in the current study is also a point of concern, as the datasets used in this study comprise a particular demographic group.

For future work, the model will be trained on more varied and difficult datasets, including videos of deepfakes, and incorporate techniques for explainability, such as Grad-CAM, which enables the visualization of the most impactful areas of the face for the model's decision. Robustness against adversarial attacks and other post-processing techniques, including JPEG compression and blurring, will also be examined, as well as extending the deployment of the model into a mobile app and exploring transformer-based hybrid architectures.

Conclusion

In this paper, we proposed an efficient 10-layer Dilated Convolutional Neural Network for real-time binary deepfake face detection. It was trained on real faces from the FFHQ dataset and fake faces from the ThisPersonDoesNotExist dataset, which were generated using StyleGAN2, to achieve a validation accuracy of 99 percent , precision of 98.85 percent , recall of 99.17 percent, and an F1-Score of 99.01 percent . The use of dilated convolutions in the later layers, which have a dilation rate of 2, helped the model detect GAN-generated artifacts over a larger spatial area without increasing the complexity of the model, thus keeping the model lightweight, requiring only 653,000 parameters.

One of the key contributions of this work is the deployment of the trained model as a real-time web application using FastAPI and React.js, which bridges the gap between research and application — a factor largely neglected in the existing deepfake detection using deep learning literature. Through the comparative analysis, it is validated that the proposed model is able to attain the same accuracy as other complex models such as XceptionNet but is the only deployed solution among the compared models.

As the development of generative AI technologies is progressing further, the gravity of the deepfake detection problem is increasing. This work has shown the feasibility of the development of accurate and deployable deepfake detection systems, making the problem more relevant in the future.

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